# ASSOCIATED POINTS IN A SHORT EXACT SEQUENCE OF QUASI-COHERENT SHEAVES

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

*Editorial note.* The proof is short: it performs the stalk reduction and cites the exact algebra analogue; the reduction is the content of the card and is stated by the author, with nothing omitted.

*Editorial note.* This card supplies the input used by the card for lemma-restriction-injective-open-contains-ass (L44225), where it enters as a navigation pointer.

*Editorial note.* Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

*Allowed standard prerequisites (editorial).*

- The algebra statement for associated primes in a short exact sequence. Algebra, Lemma algebra-lemma-ass, used at proof lines 145-147. Is the translated statement, applied at every point.

*Source reference.* Original lemma 02M3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3430–3430. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

**Lemma 0.2.** *Let $X$ be a scheme. Let $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ be a short exact sequence of quasi-coherent sheaves on $X$. Then $Ass(\mathcal{F}_2) \subset Ass(\mathcal{F}_1) \cup Ass(\mathcal{F}_3)$ and $Ass(\mathcal{F}_1) \subset Ass(\mathcal{F}_2)$.*

*Proof.* For every point $x \in X$ the sequence of stalks $0 \to \mathcal{F}_{1,x} \to \mathcal{F}_{2,x} \to \mathcal{F}_{3,x} \to 0$ is a short exact sequence of $\mathcal{O}_{X,x}$-modules. Hence the lemma follows from Algebra, Lemma 02M3. □

*Date*: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.